Welcome

ADVANCED PROJECT MANAGEMENT IN AI

Lesson 6 Session 1

Monitoring, Quality Assurance, and Case Studies:
Monitoring and Controlling

欢迎

人工智能高级项目管理

第6课 第1节

监控、质量保证与案例研究:
监控与控制

LESSON 6.1 – Monitoring and Controlling AI Projects



第6课 第1节 - 人工智能项目的监控与控制



Lesson Objectives – Lesson 6.1

- Explain why monitoring is essential for AI systems after deployment
- Identify key performance indicators (KPIs) used in AI project monitoring
- Understand model performance tracking and operational metrics
- Recognize different types of model drift and their implications
- Apply continuous improvement loops to AI systems
- Understand the role of project managers in sustaining AI system performance

课程目标 - 第6课 第1节

- 解释为何监控对于部署后的人工智能系统至关重要
- 识别用于人工智能项目监控的关键绩效指标（KPI）
- 理解模型性能跟踪和运营指标
- 识别不同类型的模型漂移（Model Drift）及其影响
- 将持续改进循环应用于人工智能系统
- 理解项目经理在维持人工智能系统性能中的作用

Why Monitoring Matters in AI Project Management

Traditional software systems are static after deployment

AI systems are dynamic and continuously evolving

Challenges for project managers:

- Data patterns change over time
- Model performance may degrade in production
- Business conditions shift
- Ethical or regulatory risks may emerge post-deployment

Therefore:

AI project success depends not only on delivery, but on sustained performance monitoring

Monitoring transforms AI from a one-time project outcome into a managed operational system

为何监控在人工智能项目管理中至关重要

传统软件系统在部署后是静态的
人工智能系统是动态的且不断演变的

项目经理面临的挑战：
- 数据模式随时间变化
- 模型性能在生产环境中可能退化
- 业务条件发生转变
- 部署后可能出现伦理或监管风险

因此：
人工智能项目的成功不仅取决于交付，还取决于持续的性能监控
监控将人工智能从一次性的项目成果转变为受管理的运营系统

Monitoring and Controlling in the AI Project Lifecycle

Monitoring occurs after deployment but throughout the operational life of the system

Key activities include:

- Tracking model accuracy and performance metrics
- Observing data inputs and distribution patterns
- Monitoring system reliability and response times
- Identifying anomalies or unexpected outcomes
- Initiating model retraining when necessary

From a PM perspective, monitoring ensures:

Alignment between model behavior and organizational objectives

人工智能项目生命周期中的监控与控制

监控发生在部署之后，贯穿系统的整个运营生命周期

关键活动包括：

•追踪模型准确性和性能指标

• 观察数据输入和分布模式

• 监控系统可靠性和响应时间

• 识别异常或意外结果

• 在必要时启动模型重新训练

从项目经理 (PM) 的角度来看，监控可以确保：

模型行为与组织目标保持一致

Key Performance Indicators (KPIs) for AI Systems

AI project KPIs can fall into three categories:

1. Model Performance Metrics

- Accuracy
- Precision / Recall
- F1 Score
- ROC-AUC

2. Operational Metrics

- Response time
- System uptime
- Prediction latency

3. Business Outcome Metrics

- Revenue impact
- Cost reduction
- User adoption
- Decision improvement

Project managers must ensure technical metrics translate into business value

人工智能系统的关键绩效指标 (KPI)

人工智能项目的 KPI 可分为三类：

1. 模型性能指标

- 准确率
- 精准率/ 召回率
- F1 分数
- ROC-AUC 曲线下面积

2. 运营指标

- 相应时间
- 系统正常运行时间
- 预测延迟

3. 业务成果指标

- 收入影响
- 成本降低
- 用户采用率
- 决策改进

项目经理必须确保技术指标转化为业务价值

Model Performance Monitoring

Model monitoring evaluates how well the AI system performs in real-world conditions

Important considerations:

- Training performance vs production performance
- Differences between test datasets and live data
- Detection of unexpected prediction behavior
- Monitoring confidence scores of predictions

Without monitoring:

A model that performed well during testing may fail silently in production

模型性能监控

模型监控评估人工智能系统在现实条件下的表现

重要考量因素：

- 训练性能和生产性能
- 测试数据集与实时数据之间的差异
- 检测意外的预测行为
- 监控预测的置信度分数

若无监控：

在测试期间表现良好的模型可能在生产环境中静默失败

Concept of Model Drift

Model drift occurs when model performance degrades over time

Main causes include:

- Changing customer behavior
- Economic or environmental shifts
- Data distribution changes
- Emergence of new patterns not present in training data

Example:

A fraud detection model trained on past transaction patterns may become less effective when new fraud strategies emerge

Drift is one of the most critical monitoring challenges in AI projects

模型漂移的概念

模型漂移是指模型性能随时间推移而退化

主要原因包括：

- 客户行为变化
- 经济或环境变化
- 数据分布变化
- 出现训练数据中不存在的新模式

示例：

基于过去交易模式训练的欺诈检测模型，当出现新的欺诈策略时，其有效性可能会降低

漂移是人工智能项目中最关键的监控挑战之一

Types of Model Drift

Data Drift

- Input data distribution changes over time
- Example: Customer demographics change

Concept Drift

- Relationship between input data and outcomes changes
- Example: Consumer behavior changes during economic recession

Prediction Drift

- Model predictions change even when input data appears stable
- Each type requires different monitoring and mitigation strategies

模型漂移的类型

数据漂移 (Data Drift)

•输入数据分布随时间变化

•示例： 客户人口统计数据发生变化

概念漂移 (Concept Drift)

•输入数据与结果之间的关系发生变化

•示例： 经济衰退期间消费者行为发生变化

预测漂移 (Prediction Drift)

•即使输入数据看似稳定，模型预测结果也发生变化

•每种类型都需要不同的监控和缓解策略

Drift Detection Techniques

Common approaches for detecting drift include:

Statistical monitoring techniques:

- Distribution comparison tests
- Population Stability Index (PSI)
- Kullback–Leibler divergence

Operational monitoring:

- Performance degradation alerts
- Threshold-based monitoring

Advanced systems may use automated drift detection pipelines

Early detection allows timely model retraining

漂移检测技术

检测漂移的常用方法包括：

统计监控技术：

- 分布比较测试
- 群体稳定性指数 (PSI)
- 库尔贝克 - 莱布勒散度

运营监控：

- 性能退化警报
- 基于阈值的监控

高级系统可能使用自动化的漂移检测管道

早期检测允许及时进行模型重新训练

Continuous Improvement Loops in AI Systems

AI systems should follow continuous learning cycles.

Typical improvement loop:

1. Monitor performance
2. Detect drift or issues
3. Collect new data
4. Retrain model
5. Validate updated model
6. Redeploy improved model

This process is often implemented through MLOps pipelines.

Continuous improvement ensures AI systems remain reliable and relevant.

人工智能系统中的持续改进循环

人工智能系统应遵循持续学习循环

典型的改进循环：

1. 监控性能
2. 检测漂移或问题
3. 收集新数据
4. 重新训练模型
5. 验证更新后的模型
6. 重新部署改进后的模型

此过程通常通过 MLOps (机器学习运维) 管道实施

持续改进确保人工智能系统保持可靠和相关性

Role of Project Managers in AI Monitoring

Project managers play a critical governance role:

Responsibilities include:

- Defining monitoring KPIs with technical teams
- Establishing monitoring schedules and review checkpoints
- Coordinating data science and operations teams
- Escalating risks when performance deteriorates
- Ensuring compliance with ethical and regulatory standards

The PM acts as the bridge between technical monitoring and business decision-making

项目经理在人工智能监控中的作用

项目经理发挥关键的治理作用；

职责包括：

- 与技术团队共同定义监控 KPI
- 建立监控时间表和审查检查点
- 协调数据科学团队和运营团队
- 当性能恶化时升级风险
- 确保符合伦理和监管标准

项目经理充当技术监控与业务决策之间的桥梁

Tools and Platforms for Monitoring AI Systems

Organizations increasingly rely on specialized monitoring tools

Examples include:

- MLflow
- Amazon SageMaker Model Monitor
- Azure Machine Learning monitoring tools
- Google Vertex AI monitoring

Capabilities include:

- automated performance tracking
- data drift alerts
- experiment tracking
- model version control

These tools enable scalable monitoring of large AI deployments

监控人工智能系统的工具与平台

组织越来越依赖专门的监控工具

示例包括：

- MLflow
- Amazon SageMaker Model Monitor
- Azure Machine Learning 监控工具
- Google Vertex AI 监控

功能包括：

- 自动化性能跟踪
- 数据漂移警报
- 实验跟踪
- 模型版本控制

这些工具实现了对大型人工智能部署的可扩展监控

Key Takeaways

AI systems require continuous monitoring after deployment.

Effective monitoring involves:

- Model performance metrics
- Operational system metrics
- Business outcome metrics

Model drift is a major challenge that must be detected early.

Successful AI projects implement continuous improvement loops.

Project managers ensure monitoring activities translate into sustained business value.

关键要点

人工智能系统在部署后需要持续监控.

有效的监控涉及：

- 模型性能指标
- 运营系统指标
- 业务成果指标

模型漂移是一个必须尽早发现的重大挑战

成功的人工智能项目实施持续改进循环

项目经理确保监控活动转化为持续的业务价值

THANKS !!!



谢谢！！！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 6 Session 2

Monitoring, Quality Assurance, and Case Studies:
Quality Assurance

欢迎

人工智能高级项目管理

第6课 第2节

监控、质量保证与案例研究:
质量保证

LESSON 6.2 – Quality Assurance in AI Projects



第6课 第2节 – 人工智能项目中的质量保证



Lesson Objectives – Lesson 6.2

- Explain why quality assurance in AI differs from traditional software QA
- Identify key validation techniques used in machine learning systems
- Understand cross-validation and testing strategies
- Evaluate the role of A/B testing in real-world AI deployment
- Explain the importance of model reproducibility and experiment tracking
- Apply documentation standards for AI governance and project transparency

课程目标 – 第6课 第2节

- 解释为何人工智能中的质量保证不同于传统软件 QA
- 识别机器学习中使用的关键验证技术
- 理解交叉验证和测试策略
- 评估 A/B 测试在现实世界人工智能部署中的作用
- 解释模型可复现性和实验跟踪的重要性
- 应用人工智能治理和项目透明度的文档标准

Why Quality Assurance Matters in AI Projects - PM Perspective

From a project management perspective, quality assurance ensures that AI systems:

- Deliver reliable and stable model performance
- Produce consistent outputs across datasets and environments
- Maintain traceability and accountability
- Support regulatory and ethical compliance
- Reduce risk of deployment failures and reputational damage

Unlike traditional systems, AI systems may degrade due to data drift, bias, or unstable training processes, making QA a continuous project governance function, not a one-time testing phase.

为何质量保证在人工智能项目中至关重要 - 项目经理视角

从项目管理的角度来看，质量保证确保人工智能系统：

- 提供可靠且稳定的模型性能
- 在不同数据集和环境中产生一致的输出
- 保持可追溯性和问责制
- 支持监管和伦理合规
- 降低部署失败和声誉受损的风险

与传统系统不同，人工智能系统可能因数据漂移、偏见或不稳定的训练过程而退化，这使得 QA 成为一项持续的项目治理职能，而非一次性的测试阶段。

Why AI Quality Assurance Is Different from Software QA

Traditional Software Testing

- Deterministic outputs
- Rule-based logic
- Stable behavior after release

AI System Quality Assurance

- Probabilistic outputs
- Performance depends on training data
- Behavior may change with new data
- Models may degrade over time

Implication for project managers:

Quality must be monitored throughout the lifecycle, not only before deployment.

为何人工智能质量保证不同于软件质量保证

传统软件测试

- 确定性输出
- 基于规则的逻辑
- 发布后行为稳定

人工智能系统质量保证

- 概率性输出
- 性能取决于训练数据
- 行为可能随新数据而变化
- 模型可能随时间退化

对项目经理的启示：

质量必须在整个生命周期中进行监控，而不仅仅是在部署之前。

Dimensions of Quality in AI Systems

Quality in AI projects must be evaluated across multiple dimensions:

Technical Quality

- Accuracy
- Precision / recall
- Model robustness

Data Quality

- Completeness
- Bias detection
- Data consistency

Operational Quality

- Latency
- Scalability
- Reliability

Governance Quality

- Transparency
- Explainability
- Auditability

AI quality assurance therefore extends beyond model accuracy alone.

人工智能系统的质量维度

人工智能项目的质量必须在多个维度上进行评估：

技术质量

- 准确率
- 精准率/ 召回率
- 模型鲁棒性

数据质量

- 完整性
- 偏见检测
- 数据一致性

运营质量

- 延迟
- 可扩展性
- 可靠性

治理质量

- 透明度
- 可解释性
- 可审计性

因此，人工智能质量保证不仅仅局限于模型准确率。

Validation Techniques in Machine Learning

Model validation ensures that the model generalizes well to unseen data.

Common validation approaches include:

Train–Test Split

- Dataset divided into training and testing portions

Holdout Validation

- Independent dataset used for final evaluation

Cross-Validation

- Multiple training/testing cycles for more reliable results

External Validation

- Model tested on completely new datasets

Project implication: Validation must be planned during project design, not after model development.

机器学习中的验证技术

模型验证确保模型能很好地泛化到未见过的数据

常用的验证方法包括：

训练 - 测试拆分

- 数据集分为训练部分和测试部分

留置验证

- 使用独立数据集进行最终评估

交叉验证

- 多次训练/测试循环以获得更可靠的结果

外部验证

- 在完全不同的新数据集上测试模型

项目启示： 验证必须在项目设计阶段规划，而不是在模型开发之后。

Cross-Validation: Ensuring Robust Model Performance

Cross-validation reduces the risk of overfitting.

Most common approach:

K-Fold Cross-Validation

Process:

1. Dataset divided into K subsets
2. Model trained on K−1 folds
3. Remaining fold used for testing
4. Process repeated K times
5. Performance averaged

Benefits:

- More reliable performance estimation
- Better use of limited datasets
- Reduced variance in evaluation results

交叉验证：确保稳健的模型性能

交叉验证降低了过拟合的风险

最常用的方法：

K 折交叉验证

过程：

1. 数据集分为 K 个子集
2. 模型在 K-1 个折上训练
3. 剩余的一个折用于测试
4. 该过程重复 K 次
5. 对性能取平均值

优势:

- 更可靠的性能估计
- 更好地利用有限的数据集
- 减少评估结果的方差

Avoiding Overfitting and Underfitting

Two major quality risks in AI systems:

Overfitting

- Model memorizes training data
- Poor performance on new data

Underfitting

- Model too simple to capture patterns

Quality assurance strategies:

- Cross-validation
- Regularization techniques
- Larger or more diverse datasets
- Feature selection

From a PM perspective, these risks influence timeline, experimentation cycles, and model reliability.

避免过拟合和欠拟合

人工智能系统中的两个主要质量风险：

过拟合

- 模型记住了训练数据
- 在新数据上表现不佳

欠拟合

- 模型过于简单，无法捕捉模式

质量保证策略：

- 交叉验证
- 正则化技术
- 更大或更多样化的数据集
- 特征选择

从项目经理的角度来看，这些风险会影响时间表、实验周期和模型可靠性。

A/B Testing in AI Deployment

Once a model is validated, real-world testing is required.

A/B testing compares two system versions:

- Version A: existing system
- Version B: new AI model

Users are randomly assigned to each version.

Performance metrics may include:

- Conversion rates
- Prediction accuracy
- Customer engagement
- Operational efficiency

A/B testing allows organizations to evaluate business impact before full deployment.

人工智能部署中的 A/B 测试

一旦模型经过验证，就需要进行现实世界测试

A/B 测试比较两个系统版本：

- 版本 A：现有系统
- 版本 B：新的人工智能模型

用户被随机分配到每个版本

性能指标可能包括：

- 转化率
- 预测准确率
- 客户参与度
- 运营效率

A/B 测试允许组织在全面部署之前评估业务影响。

Model Reproducibility in AI Projects

A critical challenge in AI projects is reproducibility.

A model should produce the same results when retrained under identical conditions.

Reproducibility requires tracking:

- Training datasets
- Data preprocessing steps
- Model parameters
- Random seeds
- Software versions
- Hardware environments

Without reproducibility, quality verification and auditing become impossible.

人工智能项目中的模型可复现性

可复现性是人工智能项目中的一个关键挑战

模型应在相同条件下重新训练时产生相同的结果

可复现性需要跟踪：

- 训练数据集
- 数据预处理步骤
- 模型参数
- 随机种子
- 软件版本
- 硬件环境

如果没有可复现性，质量验证和审计将变得不可能。

Experiment Tracking and Model Versioning

AI projects often involve hundreds of experiments.

Quality assurance requires systematic tracking of:

- Model versions
- Dataset versions
- Hyperparameters
- Performance metrics
- Training logs

Tools used in industry:

- MLflow
- Weights & Biases
- DVC
- Kubeflow

This ensures that teams can retrace decisions and reproduce results.

实验跟踪与模型版本控制

人工智能项目通常涉及数百次实验。

质量保证需要系统性地跟踪：

- 模型版本
- 数据集版本
- 超参数
- 性能指标
- 训练日志

行业使用的工具：

- MLflow
- Weights & Biases
- DVC
- Kubeflow

这确保团队可以追溯决策并重现实验结果。

Documentation Standards for AI Projects

Proper documentation is essential for governance and transparency.

Key documentation elements include:

Model Documentation

- Model purpose
- Training methodology
- Performance metrics

Dataset Documentation

- Data sources
- Collection methods
- Potential biases

System Documentation

- Deployment architecture
- Monitoring procedures
- Maintenance plans

Well-documented AI systems support regulatory compliance and stakeholder trust.

人工智能项目的文档标准

适当的文档对于治理和透明度至关重要

关键文档要素包括：

模型文档

- 模型目的
- 训练方法
- 性能指标

数据集文档

- 数据来源
- 收集方法
- 潜在偏见

系统文档

- 部署架构
- 监控程序
- 维护计划

文档完善的人工智能系统支持监管合规和利益相关者信任。

Quality Assurance Framework for AI Projects

A structured QA approach typically includes:

1. Data validation
2. Model validation
3. Deployment testing
4. Real-world performance evaluation
5. Continuous monitoring

For project managers, QA must be integrated into:

- project planning
- risk management
- governance structures
- post-deployment monitoring

AI quality assurance is therefore a continuous lifecycle process rather than a single testing phase.

人工智能项目的质量保证框架

结构化的 QA 方法通常包括：

1. 数据验证
2.模型验证
3.部署测试
4.现实世界性能评估
5.持续监控

对于项目经理，质量保证必须整合到：

- 项目规划
- 风险管理
- 治理结构
- 部署后监控

人工智能质量保证是一个持续的生命周期过程，而不是单一的测试阶段。

Key Takeaways

Quality assurance in AI projects requires:

- Validation beyond simple accuracy metrics
- Robust evaluation methods such as cross-validation
- Real-world testing using A/B experiments
- Ensuring model reproducibility
- Maintaining clear documentation and governance

Effective QA practices reduce:

- Model failure risks
- Ethical concerns
- Regulatory exposure
- Deployment instability

Quality assurance therefore plays a central role in successful AI project implementation.

关键要点

人工智能项目中的质量保证需要：

- 超越简单准确率指标的验证
- 稳健的评估方法（如交叉验证）
- 使用 A/B 实验进行现实世界测试
- 确保模型可复现性
- 保持清晰的文档和治理

有效的 QA 实践可减少：

- 模型失败风险
- 伦理担忧
- 监管暴露
- 部署不稳定性

质量保证在成功的人工智能项目实施中发挥着核心作用。

THANKS !!!



谢谢！！！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 6 Session 3

Monitoring, Quality Assurance, and Case Studies:
Case Studies

欢迎

人工智能高级项目管理

第6课 第3节

监控、质量保证与案例研究:
案例研究

LESSON 6.3 – Case Studies of AI Project Implementation



第6课 第3节 – 人工智能项目实施案例研究



Lesson Objectives – Lesson 6.3

- Analyze real-world AI implementation cases
- Evaluate AI initiatives through a project management lens
- Identify organizational and technical factors influencing AI success
- Apply lessons from industry cases to future AI project planning
- Understand how small, targeted AI projects can scale to enterprise impact

课程目标 – 第6课 第3节

- 分析现实世界的人工智能实施案例
- 从项目管理角度评估人工智能计划
- 识别影响人工智能成功的组织和技术因素
- 将行业案例的经验教训应用于未来的人工智能项目规划
- 理解小型、针对性的人工智能项目如何扩展为企业级影响

Colgate-Palmolive

Project Scope

- Generative AI used for product concept development and market research
- Employees query data, generate copy and imagery, and test concepts digitally

Implementation Approach

- AI hosted in internal hub with mandatory training
- Focused on augmenting employee creativity

Outcomes

- Faster, higher-quality concept testing
- Improved employee productivity and creativity

Source: MIT Sloan Management Review, 2025

高露洁棕榄产品

项目范围

- 利用生成式人工智能进行产品概念开发和市场调研
- 员工查询数据、生成文案和图像，并在数字环境中测试概念

实施方法

- 人工智能托管在内部中心，强制进行培训
- 专注于增强员工创造力

成果

- 概念测试更快、质量更高
- 提高员工生产力和创造力

来源：《MIT斯隆管理评论》, 2025

Colgate-Palmolive – PM Analysis

- Scope – Targeted deployment in R&D; avoids broad automation
- Stakeholders – R&D scientists, AI/data teams, leadership
- Risks – Misuse of AI outputs, adoption challenges
- Timeline & Budget – Pilot in selected teams, investment in AI tools and training
- Lessons – Small, high-impact pilots; workflow augmentation; employee engagement

高露洁棕榄产品 – 项目经理分析

- 范围： 针对研发部门的定向部署；避免广泛的自动化
- 利益相关者： 研发科学家、AI/数据团队、领导层
- 风险： 滥用 AI 输出、采用挑战
- 时间表与预算： 在选定团队中进行试点，投资 AI 工具和培训
- 经验教训： 小型、高影响力的试点；工作流程增强；员工参与

Sanofi

Project Scope

- AI guides managers to optimize investment/project decisions
- Supports decision-making, does not replace humans

Implementation Approach

- Combines generative and predictive AI for decision support
- Adjusts hierarchy of decision rights to maximize impact

Outcomes

- More informed investment choices
- Reduced sunk-cost bias

Source: MIT Sloan Management Review, 2025

赛诺菲

项目范围

- 人工智能指导管理者优化投资/项目决策
- 支持决策，不取代人类

实施方法

- 结合生成式和预测式人工智能以支持决策
- 调整决策权层级以最大化影响

成果

- 投资选择更加明智
- 沉没成本偏见减少

来源：《MIT斯隆管理评论》, 2025

Sanofi – PM Analysis

- Scope – High-value financial/project decisions
- Stakeholders – Managers, decision-makers, AI/data teams, leadership
- Risks – Over-reliance on AI, model misalignment
- Timeline & Budget – Phased deployment, investment in AI infrastructure/training
- Lessons – Human-AI collaboration is key; governance enables trusted decisions; pilots reduce risk

赛诺菲 – 项目经理分析

- 范围： 高价值的财务/项目决策
- 利益相关者： 管理者、决策者、AI/数据团队、领导层
- 风险： 过度依赖 AI、模型不一致
- 时间表与预算： 分阶段部署，投资 AI 基础设施/培训
- 经验教训： 人机协作是关键；治理赋能可信决策；试点降低风险

Liberty Mutual

Project Scope

- AI assists claims adjusters in triaging calls and resolving inquiries
- Generates multiple “choice sets” rather than a single recommendation

Implementation Approach

- AI informs human decisions, reduces cognitive load

Outcomes

- Improved call resolution efficiency
- Reduced employee cognitive load

Source: MIT Sloan Management Review, 2025

利宝相互保险公司

项目范围

- 人工智能协助理赔理算员对呼叫进行分类并解决咨询
- 生成多个“选项集”而非单一建议

实施方法

- 人工智能为人类决策提供信息，减少认知负荷

成果

- 呼叫解决效率提高
- 员工认知负荷减轻

来源：《MIT斯隆管理评论》, 2025

Liberty Mutual – PM Analysis

- Scope – Operational focus in claims handling
- Stakeholders – Claims adjusters, ops teams, AI/data teams, management
- Risks – Employee resistance, integration with systems
- Timeline & Budget – Pilot with selected teams, gradual rollout, investment in AI and training
- Lessons – AI should enhance human decision-making; pilots validate impact before scaling

利宝相互保险公司 – 项目经理分析

- 范围： 专注于理赔处理的运营
- 利益相关者： 理赔理算员、运营团队、AI/数据团队、管理层
- 风险： 员工抵触、与系统集成困难
- 时间表与预算： 在选定团队中试点，逐步推广，投资 AI 和培训
- 经验教训： 人工智能应增强人类决策；在扩展前通过试点验证影响

AI in Project Management

Generative AI and predictive AI are transforming project management workflows:

1. Planning & Scheduling – Predicts task durations, optimizes resource allocation
2. Risk Management – Flags potential risks, simulates “what-if” scenarios
3. Stakeholder & Communication Support – Auto-generates status reports, dashboards
4. Knowledge Management & Documentation – Summarizes meeting notes, maintains project repository
5. Decision Support – Prioritizes tasks/projects, aids resource trade-offs

Impact: Saves time, reduces human error, improves data-driven decisions, enhances project delivery

人工智能在项目管理中的应用

生成式人工智能和预测式人工智能正在改变项目管理的工作流程：：

1. 规划与调度： 预测任务持续时间，优化资源分配
2. 风险管理： 标记潜在风险，模拟“如果...会怎样”场景
3. 利益相关者与沟通支持： 自动生成状态报告、仪表盘
4. 知识管理与文档： 总结会议记录，维护项目存储库
5. 决策支持： 优先处理任务/项目，辅助资源权衡

影响： 节省时间，减少人为错误，改进数据驱动的决策，增强项目交付

Customer Experience Agents

- LUXGEN: AI agent reduces service workload ~30%
- Alaska Airlines: Conversational AI search for travel planning
- Volkswagen of America: myVW AI assistant with multimodal guidance
- NoBroker: Multilingual support AI handling 25–40% of future calls

Source: Google Cloud, 2024

客户体验代理

- 纳智捷: AI 代理减少约 30% 的服务工作量
- 阿拉斯加航空公司: 用于旅行规划的对话式 AI 搜索
- 美国大众集团: myVW AI 助手提供多模态指导
- NoBroker:多语言支持 AI 处理未来 25–40% 的呼叫

来源: Google Cloud, 2024

Employee Productivity & Workflow

- Uber: Gemini summarizes communications and surfaces context
- Toyota: Factory staff use AI to develop ML models, saving 10,000+ man-hours/year
- Rivian: NotebookLM centralizes answers, reducing repetitive inquiries
- Oxa: Gemini automates report templates, job descriptions, social posts

Source: Google Cloud, 2024

员工生产力与工作流

- 优步:使用 Gemini 总结沟通内容并呈现上下文
- 丰田: 工厂员工使用 AI 开发 ML 模型，每年节省 10,000+ 人工小时
- Rivian: NotebookLM 集中答案，减少重复性询问
- Oxa: Gemini 自动化报告模板、职位描述、社交媒体帖子

来源: Google Cloud, 2024

Data, Search & Operational Intelligence

- BMW Group: Vertex AI for fleet optimization and transportation analytics
- Domina (Colombia): Predicts package returns, improves delivery access by 80%
- Picterra: Scalable geospatial terrain modelling
- Teradata: Processes text/voice to categorize feedback and extract trends

Source: Google Cloud, 2024

数据、搜索与运营智能

- 宝马集团: 使用 Vertex AI 进行车队优化和交通分析
- Domina (哥伦比亚): 预测包裹退货，将配送可达性提高 80%
- Picterra: 可扩展的地理空间地形建模
- Teradata: 处理文本/语音以分类反馈并提取趋势

来源: Google Cloud, 2024

Creative Content, Marketing & Media

- Virgin Voyages: AI for hyper-personalized ads and email campaigns
- Figma: AI creates high-quality brand-approved assets in seconds
- Square Enix: Personalized marketing campaigns improve engagement/retention
- VideoShow: AI generates full short videos from simple prompts

Source: Google Cloud, 2024

创意内容、营销与媒体

- 维珍邮轮: 利用 AI 进行超个性化广告和电子邮件营销
- Figma: AI 在几秒钟内创建高质量的品牌批准资产
- 史克威尔 • 艾尼克斯 : 个性化营销活动提高参与度和留存率
- VideoShow: AI 根据简单提示生成完整的短视频

来源: Google Cloud, 2024

Closing Insight

AI success is not just about technology, it’s about people, processes, and governance.

- Small, targeted projects = measurable value = scaled adoption
- Human-AI collaboration is critical
- Data quality, workflow integration, and governance determine lasting impact

结语洞察

人工智能的成功不仅仅关乎技术，更关乎人、流程和治理。

- 小型、针对性项目 = 可衡量的价值 = 规模化采用
- 人机协作至关重要
- 数据质量、工作流集成和治理决定持久影响

Key Takeaways

1. Start small, scale smart – high-impact pilots first
2. Augment, don’t replace – AI boosts human performance
3. Governance matters – ensures adoption, trust, and compliance
4. Measurable outcomes – ROI, efficiency, and productivity are ultimate success metrics

关键要点

1. 从小处着手，智慧扩展 – 首先进行高影响力的试点
2. 增强而非取代 – 人工智能提升人类绩效
3. 治理至关重要 – 确保采用、信任和合规
4. 可衡量的成果 – 投资回报率、效率和生产力是最终成功衡量指标

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THANKS !!!



谢谢！！！

